

Differential

Homogeneous Linear Equation

OR

Cauchy - Euler Equation

A linear differential equation of the form,

$$a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_2 x^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = X$$

where, $a_0, a_1, a_2, \dots, a_{n-1}, a_n$ are the constants and X is a constant or function of x only.

Method of Solving Homogeneous Linear Differential Equation

$$\text{Let, } \log x = z \Rightarrow \frac{dz}{dx} = \frac{1}{x}$$

$$\Rightarrow x = e^z$$

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{x} \cdot \frac{dy}{dz}$$

$$\Rightarrow x \frac{dy}{dx} = \frac{dy}{dz}$$

$$\Rightarrow x \frac{d}{dx} \equiv \frac{d}{dz} = D_1$$

Also,

$$x^2 \frac{d^2 y}{dx^2} = D_1(D_1 - 1)y$$

Similarly,

$$x^3 \frac{d^3 y}{dx^3} = D_1(D_1 - 1)(D_1 - 2)y$$

$\vdots$

$$x^n \frac{d^n y}{dx^n} = D_1(D_1 - 1)(D_1 - 2) \dots (D_1 - n + 1)y$$

Solve:

$$1. \quad x^2 y_2 + x y_1 - 4 y_1 = 0$$

Solⁿ:

The given diff. eqⁿ. is,

$$x^2 y_2 + x y_1 - 4 y_1 = 0$$

$$\Rightarrow D_1(D_1 - 1)y + D_1 y - 4y = 0$$

$$\Rightarrow \{D_1(D_1 - 1) + D_1 - 4\}y = 0$$

$$\Rightarrow \{D_1^2 - D_1 + D_1 - 4\}y = 0$$

$$\Rightarrow (D_1^2 - 4)y = 0$$

The A.E. is,

$$m^2 - 4 = 0$$

$$\Rightarrow m^2 = 4$$

$$\Rightarrow m = \pm 2$$

Let, $z = \log x$

$$\Rightarrow e^z = x$$

$$C.F. = C_1 e^{2z} + C_2 e^{-2z} = C_1 e^{2 \log x} + C_2 e^{-2 \log x}$$

$\therefore$ The C.S. is,

$$y = C.F. = C_1 e^{2 \log x} + C_2 e^{-2 \log x}$$

$$2. (x^2 D^2 + 7x D + 13)y = \log x$$

Solⁿ:

Let, $z = \log x$

$$\Rightarrow e^z = x$$

$$\{D_1(D_1 - 1) + 7D_1 + 13\}y = z$$

$$\Rightarrow (D_1^2 - D_1 + 7D_1 + 13)y = z$$

$$\Rightarrow (D_1^2 + 6D_1 + 13)y = z$$

The A.E. is,

$$m^2 + 6m + 13 = 0$$

$$m = \frac{-6 \pm \sqrt{36 - 4 \times 13}}{2} = -3 \pm 2i$$

$$C.F. = e^{-3z} [C_1 \cos 2z + C_2 \sin 2z]$$

$$= e^{-3 \log x} [C_1 \cos 2 \log x + C_2 \sin 2 \log x]$$

$$= x^{-3} [C_1 \cos 2 \log x + C_2 \sin 2 \log x]$$

Now,

$$P.I. = \frac{1}{D^2 + 6D + 13} \cdot z$$

$$= \frac{1}{13 \left[1 + \frac{D^2 + 6D}{13} \right]} \cdot z$$

$$= \frac{1}{13} \left\{ 1 + \left(\frac{D^2 + 6D}{13} \right) \right\}^{-1} \cdot z$$

$$= \frac{1}{13} \left\{ 1 - \left(\frac{D^2 + 6D}{13} \right) + \left(\frac{D^2 + 6D}{13} \right)^2 - \dots \right\} \cdot z$$

Removing higher order of D ,

$$\frac{1}{13} \left[z - \frac{6}{13} \right]$$

$$\therefore P.I. = \frac{1}{13} \left(\log x - \frac{6}{13} \right)$$

Hence, the C.S. is,

$$y = C.F. + P.I.$$

$$= x^{-3} [C_1 \cos 2 \log x + C_2 \sin 2 \log x]$$

$$+ \frac{1}{13} \left(\frac{2-6}{13} \right) \cdot \underline{\text{Ans}}$$

3. $(x^2 D^2 - 3x D + 5) y = \sin(\log x)$

Solⁿ:

$$\text{Let, } z = \log x \Rightarrow e^z = x$$

$$\{D_1(D_1 - 1) - 3D_1 + 5\} y = \sin z$$

$$\Rightarrow \{D_1^2 - D_1 - 3D_1 + 5\} y = \sin z$$

$$\Rightarrow \{D_1^2 - 4D_1 + 5\} y = \sin z$$

The A.E. is,

$$\cancel{m^2} \quad m^2 - 4m + 5 = 0$$

$$m = \frac{4 \pm \sqrt{16 - 4 \times 5}}{2} = 2 \pm i$$

$$\begin{aligned}
 C.F. &= e^{2z} [C_1 \cos z + C_2 \sin z] \\
 &= e^{2 \log x} [C_1 \cos \log x + C_2 \sin \log x] \\
 &= x^2 [C_1 \cos \log x + C_2 \sin \log x]
 \end{aligned}$$

Now,

$$P.I. = \frac{1}{(D^2 - 4D + 5)} \cdot \sin z$$

$$= \frac{1}{(-1^2 - 4D + 5)} \cdot \sin z \quad [D^2 = -1^2]$$

$$= \frac{1}{(-4D + 4)} \cdot \sin z$$

$$= \frac{1}{-4(D - 1)} \cdot \sin z$$

$$= \frac{-1}{4} \cdot \frac{(D + 1)}{(D - 1)(D + 1)} \cdot \sin z$$

$$= \frac{-1}{4} \cdot \frac{(D + 1)}{(D^2 - 1)} \cdot \sin z$$

$$= \frac{-1}{4} \left(\frac{D_1(\sin z) + \sin z}{-1^2 - 1} \right)$$

$$= \frac{1}{8} (\cos z + \sin z)$$

$$= \frac{1}{8} (\cos \log x + \sin \log x)$$

Hence, the complete solution is,

$$y = C.F. + P.I.$$

$$= x^2 [C_1 \cos \log x + C_2 \sin \log x] + \frac{1}{8} \cos \log x + \frac{1}{8} \sin \log x.$$

4. $x^2 \frac{d^2 y}{dx^2} + 5x \frac{dy}{dx} + 4y = x \log x$

Solⁿ:

$$[x^2 D^2 + 5x D + 4]y = x \log x$$

Let, $z = \log x$ & $D_1 \equiv \frac{d}{dz}$
 $\Rightarrow e^z = x$

The eqⁿ. becomes,

$$[D_1(D_1 - 1) + 5D_1 + 4]y = e^z \cdot z$$

The A.E. is,

$$m(m-1) + 5m + 4 = 0$$

$$\Rightarrow m^2 - m + 5m + 4 = 0$$

$$\Rightarrow m^2 + 4m + 4 = 0$$

$$\Rightarrow (m+2)^2 = 0$$

$$\Rightarrow m = -2, -2$$

$$C.F. = (C_1 + C_2 z)e^{-2z} = (C_1 + C_2 \log x)x^{-2}$$

Now,

$$P.I. = \frac{1}{(D_1 + 2)^2} \cdot e^z \cdot z$$

$$= e^z \cdot \frac{1}{[(D_1 + 1) + 2]^2} \cdot z$$

$$= e^z \cdot \frac{1}{(D_1 + 3)^2} \cdot z$$

$$= e^z \cdot \frac{1}{9\left(1 + \frac{D_1}{3}\right)^2} \cdot z$$

$$= \frac{e^z}{9} \left[1 + \frac{D_1}{3}\right]^{-2} \cdot z$$

$$= \frac{e^z}{9} \left[1 - \frac{2D_1}{3} + 3\left(\frac{D_1}{3}\right)^2 - \dots\right] \cdot z$$

Removing higher order of D_1 ,

$$\frac{e^z}{9} \left[1 - \frac{2D_1}{3}\right] z$$

$$= \frac{e^z}{9} \left(z - \frac{2}{3}\right)$$

$$= \frac{e^z}{27} (3z - 2)$$

$$\therefore \text{P.I.} = \frac{x}{27} (3 \log x - 2)$$

Hence, the complete solution is,

$$y = \text{C.F.} + \text{P.I.}$$

$$= (C_1 + C_2 \log x) x^{-2} + \frac{x}{27} (3 \log x - 2)$$

$$5. \quad \frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} = \frac{12 \log x}{x^2}$$

Solⁿ:

Multiplying the given eqⁿ. by x^2 , we get,

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = 12 \log x$$

$$\text{Let, } z = \log x$$

$$\Rightarrow e^z = x$$

$$\& \quad D_1 \equiv \frac{d}{dz}$$

$$[D_1 (D_1 - 1) + D_1] y = 12 z$$

The A.E. is,

$$m(m-1) + m = 0$$

$$\Rightarrow m^2 - m + m = 0$$

$$\Rightarrow m = 0, 0$$

$$C.F. = (C_1 + C_2 z) = (C_1 + C_2 \log x)$$

Now,

$$P.I. = \frac{1}{D_1^2} \cdot 12z$$

$$= \frac{12}{D_1} \int z \, dz$$

$$= 12 \int \frac{z^2}{2} \, dz$$

$$= 6 \times \frac{z^3}{3}$$

$$= 2z^3$$

$$= 2(\log x)^3$$

Hence, the complete solution is,

$$y = C.F. + P.I.$$

$$= (C_1 + C_2 \log x) + 2(\log x)^3$$

$$6. x^2 D^2 y - 3x Dy + 5y = x^2 \sin \log x$$

Solⁿ:

$$\text{Let, } z = \log x \quad \& \quad D_1 \equiv \frac{d}{dz} \equiv x \frac{d}{dx}$$

$$\Rightarrow e^z = x$$

$$[D_1(D_1 - 1) - 3D_1 + 5]y = e^{2z} \sin z$$

The A.E. is,

$$m(m-1) - 3m + 5 = 0$$

$$\Rightarrow m^2 - m - 3m + 5 = 0$$

$$\Rightarrow m^2 - 4m + 5 = 0$$

$$\Rightarrow m = \frac{4 \pm \sqrt{16 - 4 \times 5}}{2} = 2 \pm i$$

$$\text{C.F.} = e^{2z} [C_1 \cos z + C_2 \sin z]$$

$$= e^{2 \log x} [C_1 \cos \log x + C_2 \sin \log x]$$

Now,

$$\text{P.I.} = \frac{1}{(D_1^2 - 4D_1 + 5)} \cdot e^{2z} \sin z$$

$$= e^{2z} \cdot \frac{1}{[(D_1+2)^2 - 4(D_1+2) + 5]} \cdot \sin z$$

$$= e^{2z} \cdot \frac{1}{(D_1^2 + 4D_1 + 4 - 4D_1 - 8 + 5)} \cdot \sin z$$

$$= e^{2z} \cdot \frac{1}{(D_1^2 - 4 + 1)} \cdot \sin z$$

$$= e^{2z} \cdot \frac{1}{-1^2 + 1} \cdot \sin z$$

Case of failure,

$$= z e^{2z} \cdot \frac{1}{2D_1} \cdot \sin z$$

$$= \frac{z e^{2z}}{2} \int \sin z \, dz$$

$$= -\frac{z e^{2z}}{2} \cdot \cos z$$

$$= -\frac{x^2}{2} \log x \cdot \cos \log x$$

Hence, the C.S. is,

$$y = e^{2 \log x} [C_1 \cos \log x + C_2 \sin \log x] - \frac{x^2}{2} \log x \cos \log x.$$

Equations Reducible to Homogeneous Linear Differential Equation

If any equation of form, which is also called Legendre's Linear Equation, then it is reduced to homogeneous linear differential equation.

Now, the Legendre's linear equation is,

$$[a_0(ax+bx)^n D^n + a_1(ax+bx)^{n-1} D^{n-1} + \dots + a_{n-1}(ax+bx) D + a_n] y = X \quad (1)$$

where, $a_0, a_1, \dots, a_n$ are constants. •

Solution of Differential Eqⁿ. (1):

Note: Consider $(ax+b)$ as $(a+bx)$.

$$\text{Let, } \log(ax+b) = z \\ \Rightarrow ax+b = e^z$$

$$\frac{d}{dx} \equiv D \quad \& \quad x \frac{d}{dz} \equiv D_1$$

$$(ax+b) \frac{d}{dz} \equiv a D_1$$

$$(ax+b)^2 \frac{d^2}{dz^2} \equiv a^2 D_1(D_1 - 1)$$

& so on.

Solve:

$$1. (1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + y = 4 \cos \log(1+x)$$

Solⁿ:

$$\text{Let, } \log(1+x) = z \\ \Rightarrow 1+x = e^z$$

$$[1^2 D_1 (D_1 - 1) + 1 \cdot D_1 + 1] y = 4 \cos z$$

$$\Rightarrow (D_1^2 + 1) y = 4 \cos z$$

The A.E. is,

$$m^2 + 1 = 0$$

$$\Rightarrow m^2 = -1$$

$$\Rightarrow m = \pm i$$

$$\text{C.F.} = C_1 \cos z + C_2 \sin z = C_1 \cos \log(1+x) + C_2 \sin \log(1+x)$$

$$\text{P.I.} = \frac{1}{(D_1^2 + 1)} \cdot 4 \cos z$$

$$= 4 \cdot \frac{1}{-1^2 + 1} \cos z$$

Case of failure,

$$\text{P.I.} = 4z \cdot \frac{1}{2D_1} \cdot \cos z$$

Date _____

$$= 2z \int \cos z \, dz$$

$$= 2z \sin z$$

$$= 2 \log(1+x) \cdot \sin[\log(1+x)]$$

Hence, the complete solution is,

$$y = C.F. + P.I.$$

$$= C_1 \cos \log(1+x) + C_2 \sin \log(1+x) + 2 \log(1+x) \cdot \sin[\log(1+x)] \quad \underline{\text{Ans}}$$

Date _____

$$2. \{(x+1)^4 D^3 + 2(x+1)^3 D^2 - (x+1)^2 D + (x+1)\}y = \frac{1}{x+1}$$

Solⁿ:

The given eqⁿ. can be written as,

$$\{(x+1)^3 D^3 + 2(x+1)^2 D^2 - (x+1)D + 1\}y = (1+x)^{-2} \quad \text{--- (1)}$$

Let, $z = \log(x+1)$

$$\Rightarrow e^z = x+1$$

$$\Rightarrow x = e^z - 1$$

$$\{D_1(D_1-2)(D_1-1) + 2D_1(D_1-1) - D_1 + 1\}y = e^{-2z}$$

$$\Rightarrow \{D_1(D_1^2 - D_1 - 2D_1 + 2) + 2D_1^2 - 2 - D_1 + 1\}y = e^{-2z}$$

$$\Rightarrow \{D_1^3 - D_1^2 - 2D_1^2 - 2D_1 + 2D_1^2 - 2 - D_1 + 1\}y = e^{-2z}$$

$$\Rightarrow [D_1^3 - D_1^2 - D_1 + 1]y = e^{-2z}$$

The A.E. is,

$$m^3 - m^2 - m + 1 = 0$$

$$\Rightarrow m^2(m-1) - (m-1) = 0$$

$$\Rightarrow (m^2-1)(m-1) = 0$$

$$\Rightarrow m^2 = 1, \quad m = 1$$

$$\Rightarrow m = \pm 1, 1$$

$$C.F. = (C_1 + C_2 z)e^z + C_3 e^{-z}$$

$$= \{C_1 + C_2 \log(x+1)\}(x+1) + C_3(x+1)^{-1}$$

$$P.I. = \frac{(x+1)^{-1} \cdot e^{-2x}}{(D^3 - D^2 - D + 1)}$$

$$= \frac{e^{-2x}}{(-2)^3 - (-2)^2 - (-2) + 1}$$

$$= \frac{e^{-2x}}{-8 - 4 + 2 + 1}$$

$$= \frac{-e^{-2x}}{9}$$

$$= \frac{-(x+1)^{-2}}{9}$$

Hence, the C.S. is,

$$y = C.F. + P.I.$$

$$= \{C_1 + C_2 \log(x+1)\}(x+1) + C_3(x+1)^{-1} - \frac{(x+1)^{-2}}{9}$$

$$3. [(1+2x)^2 D^2 - 6(1+2x)D + 16] y = 8(1+2x)^2$$

Solⁿ:

$$\text{Let, } \log(1+2x) = z$$

$$\Rightarrow 1+2x = e^z$$

$$[2^2 D_1 (D_1 - 1) - 6 \times 2 D_1 + 16] y = 8e^{2z}$$

$$\Rightarrow [4D_1^2 - 4D_1 - 12D_1 + 16] y = 8e^{2z}$$

$$\Rightarrow [4D_1^2 - 16D_1 + 16] y = 8e^{2z}$$

$$\Rightarrow 4[D_1^2 - 4D_1 + 4] y = 8e^{2z}$$

$$\Rightarrow (D_1^2 - 4D_1 + 4) y = 2e^{2z}$$

The A.E. is,

$$m^2 - 4m + 4 = 0$$

$$\Rightarrow m = \frac{-(-4) \pm \sqrt{(-4)^2 - 4 \times 1 \times 4}}{2 \times 1} = \frac{4 \pm 0}{2} = 2, 2$$

$$\Rightarrow m = 2, 2$$

$$C.F. = (C_1 + C_2 z) e^{2z}$$

$$= (C_1 + C_2 \log(1+2x)) (1+2x)^2$$

$$P.I. = \frac{1}{(D_1^2 - 4D_1 + 4)} \cdot 2e^{2z}$$

$$= 2e^{2z} \cdot \frac{1}{2^2 - 2 \times 4 + 4}$$

Case of failure,

$$P.I. = 2z \cdot \frac{1}{2D_1 - 4} \cdot e^{2z}$$

Case of failure

$$P.I. = 2z^2 \cdot \frac{1}{2} \cdot e^{2z}$$

$$= z^2 e^{2z}$$

$$= [\log(1+2x)]^2 (1+2x)^2$$

Hence, the C.S. is,

$$y = C.F. + P.I.$$

$$\therefore y = [C_1 + C_2 \log(1+2x)] (1+2x)^2 + \frac{[\log(1+2x)]^2}{(1+2x)^2}$$

Ans